

ELEC5305 Project Proposal

Audio Noise Reduction Using STFT-Based Spectral Processing

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GitHub Repository:

<https://github.com/kaiqi-yuan/elec5305-project-530049256>

GitHub Pages:

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1 Project Overview

Audio recordings are often affected by unwanted background noise from the environment, such as street sounds, room reverberation, air conditioning systems, and electronic devices. This noise can degrade the quality and intelligibility of speech and other audio signals. This project aims to develop an audio denoising system based on spectral processing techniques utilizing the short-time Fourier transform (STFT).

The core idea is to use STFT to convert noisy audio signals from the time domain into a time-frequency representation, estimate the spectral characteristics of the background noise, and suppress unwanted spectral components. After processing, the modified spectrum is then converted back into a time-domain signal using the inverse STFT.

The project is primarily implemented in MATLAB. System performance will be evaluated through waveform comparisons, spectrograms, signal-to-noise ratio (SNR), and subjective listening tests. The ultimate goal is to build a functional prototype capable of effectively reducing background noise while preserving the essential components of the desired audio signal.

2 Background and Motivation

Noise reduction is an important application in digital signal processing, widely used in voice recording, mobile communication, video conferences, hearing aids, and speech recognition systems. In actual recording environments, microphones usually pick up both the desired signal and unwanted background noise simultaneously. Therefore, signal processing techniques are needed to remove these unwanted components while minimizing any significant distortion to the desired audio.

Fourier analysis can be used to identify the frequency components in audio signals. However, a single Fourier transform analyzes the entire signal duration, which cannot directly reflect the changes in its frequency components over time. This is an important limitation for signals such as speech and music, as their spectral characteristics are constantly changing.

The Short-Time Fourier Transform (STFT) solves this problem by analyzing short segments of the signal. The audio signal is divided into shorter and usually overlapping frame segments. A window function is applied to each frame. By repeating

this process over the entire signal, a time-frequency representation can be obtained, which can be visualized using a spectrogram.

This project was selected because it covers the practical application of multiple concepts learned in the ELEC5305 course, including Fourier analysis, windowing processing, spectral analysis, and the Short-Time Fourier Transform (STFT). Additionally, this project also provides an opportunity to study how different signal processing parameters affect the final audio output quality.

3 Proposed Methodology

The project will begin with several clean audio recordings, primarily speech signals. Different types and levels of background noise will be added to the clean recordings to create controlled noisy test signals. This approach allows the original clean signal to remain available as a reference when evaluating the performance of the noise reduction algorithm.

The noisy audio signal will first be divided into short overlapping frames. A window function, such as a Hann window, will be applied to each frame to reduce discontinuities at the frame boundaries. The STFT will then be used to obtain a complex time-frequency representation of the signal.

A discrete STFT can be represented as

$$X(m, k) = \sum_{n=0}^{L-1} x[n + mH] w[n] e^{-j2\pi kn/N}, \quad (1)$$

where L represents the frame length, H is the hop size, $w[n]$ is the window function, N is the FFT size, m represents the frame index and k represents the frequency-bin index.

The background noise spectrum will then be estimated from noise-dominant sections of the recording. The estimated noise spectrum will be used to reduce unwanted spectral components. The initial implementation will use spectral subtraction or spectral attenuation because these methods provide a clear demonstration of frequency-domain noise reduction.

After spectral processing, the enhanced audio signal will be reconstructed using the inverse STFT and overlap-add processing. Different parameters, including frame length, hop size, FFT size and noise reduction strength, will be tested to investigate their effects on the final result.

If sufficient time is available, a Wiener-filter-based method will also be implemented. Its performance will be compared with the initial spectral subtraction method.

4 Expected Outcomes

The expected outcome of this project is a working MATLAB-based audio noise reduction prototype. The system should be able to accept a noisy audio recording, analyse the signal in the time-frequency domain, reduce estimated background noise and reconstruct an enhanced audio signal.

The final project is expected to provide:

- original, noisy and enhanced audio examples;
- waveform comparisons before and after processing;
- spectrogram comparisons of noisy and enhanced signals;
- SNR measurements before and after noise reduction;
- analysis of different STFT and noise reduction parameters; and
- MATLAB source code and project documentation hosted on GitHub.

The performance of the system will be evaluated both quantitatively and qualitatively. Where a clean reference signal is available, the SNR before and after processing will be calculated. Listening tests and spectrogram comparisons will also be used to identify audible improvements or unwanted distortion.

A successful result will demonstrate measurable noise reduction while maintaining acceptable audio quality. The project will also investigate limitations of spectral noise reduction, particularly when the frequency content of the background noise overlaps with the desired signal.

5 Project Timeline

Weeks	Task
Weeks 1–2	Project topic selection and initial planning.
Weeks 3–5	Literature review and collection of clean and noisy audio samples.
Weeks 6–7	Implementation of STFT analysis and spectrogram visualisation in MATLAB.
Weeks 8–9	Implementation of noise estimation and spectral noise reduction.
Weeks 10–11	Parameter testing, SNR evaluation and comparison of results.
Week 12	Final optimisation, testing and analysis of limitations.
Week 13	Final report, MATLAB code organisation and GitHub documentation.

6 References

1. Boll, S. F. (1979). Suppression of acoustic noise in speech using spectral subtraction. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 27(2), 113–120.
2. Ephraim, Y., and Malah, D. (1984). Speech enhancement using a minimum mean-square error short-time spectral amplitude estimator. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 32(6), 1109–1121.
3. Griffin, D. W., and Lim, J. S. (1984). Signal estimation from modified short-time Fourier transform. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 32(2), 236–243.
4. Loizou, P. C. (2013). *Speech Enhancement: Theory and Practice*. CRC Press.

7 GitHub Project

All MATLAB source code, audio samples, experimental results and project documentation will be maintained in a public GitHub repository.

GitHub Repository:

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The repository contains the project proposal and README documentation. Additional MATLAB source code, audio samples and experimental results will be added as the project develops.